

Date: Nov 27, 2025
To: "Gregory Paradis" gregory.paradis@ubc.ca
From: "Ecological Informatics" support@elsevier.com
Subject: Decision on submission to Ecological Informatics
 **Attachment(s):** [Review final.docx](#)

Manuscript Number: **ECOINF-D-25-03829**

WS3: An open-source Python framework for integrated simulation and optimization of forest landscape and wood supply systems

Dear Dr. Paradis,

Thank you for submitting your manuscript to Ecological Informatics.

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following major revision. I invite you to resubmit your manuscript after addressing the comments below. **Please see also attached report!** Please resubmit your revised manuscript by **Jan 11, 2026**.

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed.

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Ecological Informatics values your contribution and I look forward to receiving your revised manuscript.

Kind regards,
George Arhonditsis
Editor-in-Chief

Ecological Informatics

Editor and Reviewer comments:

Reviewer #2: In general, the work of this research group seems quite sophisticated, and their models seem elegant in design. However, I have mixed feelings about the work as it is presented in this manuscript - is it science or is it a sales pitch? The description of the modeling system goes a bit too far down the "sales-pitch" route with terms such as "exposes" and "ships" and "releases the tool chain." In certain places (e.g., lines 100, 103, 155, 167) these terms seem obtuse. Further, the assignment of certain adjectives ("first class" on line 361, "easy" on line 364) are value-driven and not an outcome of science.

A further concern is that the claims in the abstract may be too strong. The validation benchmark against Woodstock seems to have been attempted using one hypothetical (toy) dataset. The broader forest planning models (and the hypothetical) are not described sufficiently to get a sense of the case studies, and only seem useful for the "scaling experiments."

The modeling approaches are not fully described either. My interpretation of the work is that a strata-based LP problem is solved, and that a heuristic search process is employed to allocate the outcomes back to an actual forest. Perhaps I am wrong, which further underscores a need to be clearer about the problem-solving systems described in the manuscript.

The LP model is a strata-based Model I, which should be a little more explicit in Section 2.2. In this case the stand identities are lost as they are combined into a stratum. Perhaps the heuristic is used to take the LP results and allocate them back to the ground, as the Stanley model attempted to do with Woodstock outcomes. If so, some mention of Stanley would be appropriate when describing the heuristic.

A few other thoughts about the heuristic search process are unclear. Is the heuristic search process an integer model? It appears that it works in raster space, informed by the proportions of strata harvested in the LP results. How is the area control method employed with the heuristic search process? How are the actions scheduled with the heuristic? Is it a bin packing algorithm that schedules activities up to some pre-defined area or volume target, or is it more like binary search where the target is initially unknown, and some even level of output is desired? In either case, it seems to want to operate like binary search in the sense that an "oldest first" rule is employed, which can only be accomplished sufficiently if activities are scheduled one time period at a time, from 1-n. Some context in comparison with other methods such as these would be helpful.

Minor issues

Line 353: "codebases"
 Line 366: "decisiveeven"
 Line 381: "pressuretens"
 Line 382: "benchmarksbecause"
 Line 509: Forest Science. 23(1): Monograph 18.

Reviewer #3: The study documents the modular software architecture of WS3, a decision support system (DSS) designed for harvest scheduling with integrated carbon bookkeeping within a raster-based allocation framework. It further compares WS3's performance against industrial standards through computational scaling experiments and provides the complete toolchain, including source code, configuration files, and deterministic reproduction scripts. The paper demonstrates the potential of WS3 to enable agencies, industry, Indigenous governments, and researchers to co-develop climate-aligned harvest strategies, evaluate fuel management and nature-based solution scenarios, and integrate harvest scheduling with broader ecological simulations.

While the manuscript is clearly written and presents its results in a coherent manner, it does not provide substantial or novel advancements in the field of DSS or forest management planning when compared with well-established tools such as WOODSTOCK, Patchworks, JLP (MELA), and Heureka. Several key aspects therefore require further clarification, methodological justification, and elaboration to strengthen the contribution and originality of the work.

Major Comments

1. Methodological depth and innovation

The main shortcomings of the study stem from the insufficient description and elaboration of the methodology used to develop the open-source DSS for harvest scheduling and carbon accounting. The paper lacks a clear methodological framework and fails to articulate how WS3 advances the current state of the art. Specifically, it does not introduce any novel approaches for:

- * addressing risks and uncertainty,
- * integrating multiple ecosystem services,
- * solving spatial optimization problems,
- * incorporating artificial intelligence,
- * coupling with process-based modeling, or
- * embedding digitalization/virtualization concepts into DSS design and implementation.

Furthermore, the study provides little methodological detail on how management objectives and strategies are formulated or improved compared to existing systems. This lack of methodological rigor makes it difficult to assess the scientific novelty or practical significance of WS3. Without such clarity, the conceptual foundation of the research remains vague, and its empirical contributions appear limited and insufficiently substantiated.

2. Introduction and conceptual framing

The Introduction section includes an excessive amount of technological jargon, which obscures the main scientific rationale and methodological intent of the study. Instead of focusing on technical details, the introduction should clearly articulate the conceptual foundation, research motivation, and practical relevance of the work. The current version lacks a coherent framework that connects technological developments with the broader challenges of DSS design for sustainable forest management.

The authors should elaborate on the theoretical basis and rationale for developing a "new era" or "new frontier" DSS—explicitly identifying the existing challenges, knowledge gaps, and opportunities their approach seeks to address. Such contextualization is crucial for readers to understand the significance, novelty, and broader implications of the study.

3. Demonstration of model advantages

The study must more clearly demonstrate the advantages of WS3 in terms of its architecture, computational efficiency, and functional capabilities relative to existing DSS. The manuscript describes five core components—(i) core data abstractions, (ii) forest model construction and simulation, (iii) optimization, (iv) spatial allocation, and (v) shared utilities—but these features are already common in most established systems, including those referenced in the paper.

Moreover, many existing tools already handle spatial operational constraints such as adjacency and opening size, which WS3 does not appear to address effectively. Consequently, the claimed novelty and strengths of WS3 remain unsubstantiated. The authors should explicitly identify the unique and critical functionalities that distinguish WS3 from comparable systems—particularly features that other DSS platforms do not support or perform less efficiently. Without this differentiation, it is difficult to recognize the model's genuine contribution to forest management planning and DSS innovation.

4. Conclusion and synthesis

The Conclusion section does not effectively communicate what has been accomplished through the study. Instead, it reads as a loosely structured summary that reiterates technological details without presenting the key findings or scientific contributions in a concise and integrative manner. The section should focus on synthesizing the main achievements, demonstrating the added value of WS3, and highlighting the implications for future DSS development and forest management applications.

Additionally, the extended discussion on future work is unnecessary, as limitations have already been mentioned earlier in the manuscript. A brief acknowledgment of these limitations would suffice. Overall, the conclusion resembles a condensed version of a technical report or thesis documentation of software development rather than a focused synthesis that reinforces the study's main outcomes and broader relevance.

Minor Comments

* Programming-level detail: Certain programming-related descriptions, such as "Listing 1: Case-study workflow executed by the reproduction package," appear unnecessary in a scientific article. These elements add technical complexity without enhancing conceptual clarity and may not be meaningful for most readers. Simplifying or omitting such details would improve readability and focus.

* Definition of decision variables: The definition of the key decision variable $X_{i,j}$ lacks clarity. The authors should explicitly specify what $X_{i,j}$ represents—whether it denotes age or size classes, stand types, individual stands, or spatial cells. The current explanation, "the fraction of zone i assigned to prescription j ," leaves the meaning of "zone" ambiguous. It is unclear whether a zone refers to an aggregation of stands, grid cells, or stand types. A clearer and more consistent spatial definition is essential to ensure that the decision variables and their ecological or management implications are conceptually sound and methodologically transparent.

Overall Assessment

The manuscript has merit in promoting open-source DSS development and reproducible research in forest management. However, it requires significant revision to clarify its methodological framework, articulate its scientific novelty, and more convincingly demonstrate the practical advantages of WS3 relative to existing systems. Addressing the above concerns would substantially enhance the rigor, readability, and overall contribution of the paper.

Reviewer #4: The proposed implementation builds upon existing work and the objective function does not really meet the ecological scope of the journal. To this reviewer, the work is more designed for operational research journals than for ECOINF.

Moreover, the novelties from this work are minimal with respect to problem formulation, innovative constraints, etc. I agree it make it easier for users to operate a DSS tool but beyond that the contribution to the state of the art is minimal. I strongly recommend to explore a journal for prone for operational research.

Reviewer #5: Manuscript ECOINF-D-25-03829 "WS3: An open-source Python framework for integrated simulation and optimization of forest landscape and wood supply systems" - the Reviewer's comments.

The manuscript describes the WS3 Python software that links the outputs of a harvest planning model based on a simple harvest scheduling problem I with the existing forest Carbon Budget Model (libCBM) to calculate the forest carbon stocks for the harvest planning solutions generated with the harvest scheduling model. The manuscript claims to make three science contributions, which include presenting the new WS3 software and validating the software against the commercial Woodstock harvest planning software and through computational experiments. I believe these contributions may qualify as software development advancements but not as notable carbon / forest management or information science contributions.

The WS3 software makes it easier to calculate the forest carbon stocks with the Canadian Forest Service Carbon Budget Model (libCBM) for the spatial harvest prescriptions generated with the most basic LP harvest scheduling problem I. The software is designed to improve handling of forest inventory datasets (which are the standard inputs for harvest scheduling calculations and forest carbon accounting). However, the presented software does not incorporate carbon accounting into an optimization-based linear programming framework. The presented harvest scheduling model does not consider carbon

in the objective function or the model constraints; the carbon balance estimates are calculated post-hoc, after the timber-only harvest solution is found. In theory, one could undertake similar analyses by running the harvest plans generated with a commercial harvest planning software through the CBM3 Carbon Budget Model.

The software would provide a reasonable science contribution if it could allow formulating the forest carbon management objectives (or constraints) as part of the harvest scheduling model I. This would make the software useful for exploring the carbon retention trade-offs in forest management planning. However, the software does not offer this option and applies basic harvest scheduling problem I formulation without carbon.

The text has a noticeable imprint of LLM-based text editing tools and should be revised to reduce the excessive use of jargon and subheadings and make the narrative easier to understand for readers with limited forest management planning background. The manuscript style resembles a research grant report that describes the software and the model runs without providing much background about the science of integrating forest carbon accounting into an optimization-based forest planning framework. There is no literature review of previous applications of forest carbon accounting in optimization-based forest planning models.

Overall, the manuscript describes a new environmental software but fails to provide a major contribution to forest management planning and information science. The paper, after an editorial overhaul, could be a reasonable submission to a journal that publishes the descriptions of new software and analysis protocols.

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